HuskySort

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Most sorting algorithms in the literature optimize for the number of comparisons or exchanges; we argue the more appropriate yardstick is the total number of array accesses (the "work"), which for divide-and-conquer sorts splits into a linear, $\mathbf{O}(N)$, phase and a linearithmic, $\mathbf{O}(N \log N)$, phase. Moving work out of the linearithmic phase and into the linear phase reduces this total and, where comparison itself is expensive, reduces processing time as well. The key concept is a 64-bit code that is, as far as possible, order-preserving, and stands in as a proxy for each original element; we call it a "Husky" code. Its effect is to extract each element's sort key once, rather than once per comparison. We present two algorithms built on it. QuickHuskySort encodes each object in a linear preamble, sorts the cheap encoded keys in place of the expensive originals, and falls back to a cleanup pass only where the encoding is imperfect. RadixHuskySort replaces that sort with a linear-time radix sort on the same keys, moving the second phase itself from linearithmic to linear. Measured against Java's system sort for objects — across integers, arbitrary-precision numbers, dates, tuples, English and Chinese text, Chinese personal names in pinyin order, and two hundred thousand municipal records ordered by a composite key — RadixHuskySort is faster in every case, by 3.6–6.8x; QuickHuskySort, which retains a comparison sort, by 1.4–2.5x wherever comparison is genuinely costly. The clearest demonstration is the pinyin ordering, where obtaining a single character's sort key requires a table lookup. Compared against a system sort performing that same ordering — rather than the default one, which sorts by code point and so answers a different and much cheaper question — RadixHuskySort is faster by 4.1x at one million names and QuickHuskySort by 2.5x, and the margin widens as the array grows. That widening is the premise made visible: what is saved is one key extraction per element in place of one per comparison. The margin is widest of all when the ordering is expensive to evaluate and the encoding is also exact, so that no cleanup pass is needed: measured on input where that pass provably has nothing to correct, it nonetheless accounts for as much as a quarter of total running time. It is narrowest where algorithms specialised to the type already exist — on English words a tuned MSD radix sort matches or overtakes us — which bounds the result rather than contradicting it, since the same mechanism applies unchanged to types for which no specialised sort exists.

CCS Concepts: • **Theory of computation → Sorting and searching**; **Data structures and algorithms for data management**; • **Software and its engineering → Data types and structures**; • **Computer systems organization** → Multicore architectures;

Additional Key Words and Phrases: Sorting, Quicksort, Timsort, Radix sort, Complexity, Comparison Sorting, String sorting, Parallel sorting

1 INTRODUCTION

With the vast increase in the size of the data world, efficient computing algorithms play a crucial role in processing data. From the earliest days of computing, sorting has been one of the most studied areas of research even though it appears to be a straightforward and familiar topic. The rise of data volume and complexity has made clear a need for new approaches to traditional sorting techniques. Within the realm of comparison-sort algorithms, many aspects of the algorithm can be significant: stability, adaptability, extra memory, partitioning, merging, exchanging (swapping), copying, and, of course, the comparison itself.

Counting memory operations rather than only comparisons is itself an established idea, and we frame our own array-access model relative to two well-known lines of work in that tradition rather than proposing it as new: the external-memory (I/O) model of Aggarwal and Vitter [Aggarwal and Vitter 1988], which counts block transfers between disk and memory, and the cache-oblivious algorithms of Frigo et al. [Frigo et al. 1999], which achieve near-optimal memory-hierarchy behaviour without advance knowledge of cache parameters. Our array-access count is a simpler, single-level analogue of the same underlying idea: rather than modeling an explicit multi-level memory hierarchy, we count raw array reads and writes as a proxy for real cost, motivated by the observation that a comparison between two heavyweight objects (e.g., two *String*s) touches more memory — and likely non-cached memory — than a comparison between two 64-bit primitives. That observation is orthogonal to, not a substitute for, the external-memory and cache-oblivious literature, which is chiefly concerned with data too large to fit in memory (or in cache) at all, rather than with the comparison cost of individual elements that do fit. Separately, radix sort's independent per-digit counting-sort passes are well suited to parallelization across CPU threads, a property that is already established for radix sort generally; § 6.4 describes and benchmarks a parallel variant of RadixHuskySort (§ 3.2). We do not make an equivalent claim for HuskySort's own comparison-based steps: Introsort's partitioning and Timsort's adaptive run-detection are both substantially more data-dependent (less independent) and sequential in character than radix sort's fixed per-digit passes, so parallelizing them well is a separate, harder problem that this paper does not address.

Even though sorting appears to be a settled area of research, yet new improvements are still being made: [Bentley and McIlroy 1993; Jugé 2020; McIlroy 1993; Peters 2002; Yaroslavskiy 2009]. Consequently, there is an abundance of sorting techniques available for a myriad of different use cases. Nevertheless, there is still room for improved versions in regard to computation, memory, time, and space complexity. In a recent paper[Zhang et al. 2016], the authors

*originator of the idea; did the analysis; designed, coded, and performed much of the benchmarking. The developed HuskySort Code is available at: https://github.com/rchillyard/HuskySort

†implemented the initial algorithm, including the coders and the key innovation of the object swap in the quicksort phase; early benchmarking.

‡Analysed the data, extended the literature survey and wrote the paper with input from all authors.

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conclude that quicksort is indeed the fastest general-purpose sorting algorithm, especially when appropriate care is taken to avoid the quadratic worst-case with several other improvements (e.g., dual-pivot quicksort)[Yaroslavskiy 2009] over the traditional version[Hoare 1961]. However, quicksort does not perform as well as some variations of merge sort, especially Timsort[Peters 2002], when the dataset is partially sorted. Timsort, a hybrid stable sorting algorithm derived from merge sort, is now available in the Python and Java libraries as the default object-sorting utility.

One of the authors had originally posed this as a hypothetical question in a mid-term exam (Fall 2018): encode a date-time object as a 64-bit hash code and sort the array of codes with quicksort. It turned out to be a pretty good idea, and we decided to test the efficacy of combining quicksort with Timsort for object types that are costly to compare. Such types include *String, LocalDateTime, BigInteger, BigDecimal*, and many user-defined types of this kind.

This paper builds on the work of Hillyard et al.[Hillyard et al. 2020], which introduced the husky code and the QuickHuskySort algorithm that sorts by it. The contributions here are RadixHuskySort (§ 3.2), which replaces that algorithm's comparison sort with a linear-time radix sort on the same codes, and a parallel variant of it (§ 6.4); an empirical comparison against the classic string sorts rather than a theoretical one (§ 3.2); a direct measurement of the cost of the cleanup pass (§ 3.4); and a case study on real rather than generated data (§ 6.3). All results reported here were measured anew.

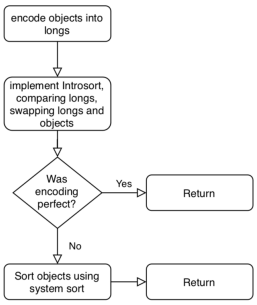
We review existing methods of sorting in Section 2 and go through the HuskySort algorithm in Section 3. Implementation options and their likely impact are explored in Section 4. We describe our Test Cases in Section 5 and present results in Section 6, with outcomes summarized in Section 7.

2 BACKGROUND

Before introducing HuskySort, it is worth laying out the four comparison-based algorithms it draws on side by side, since each contributes a different property to the final design. Table 1 summarizes their standard complexity and stability characteristics.

Quicksort is fastest in practice on unordered data (its actual average case), because its in-place partitioning and cache-friendly access pattern outweigh merge sort's guaranteed $\mathbf{O}(N \log N)$ bound; but it buys that speed at the cost of stability, and its worst case is quadratic unless mitigated (as Introsort does, by falling back to heapsort past a recursion-depth threshold — see § 3.2 below). Timsort takes the opposite trade: guaranteed $\mathbf{O}(N \log N)$ worst case and stability, at the cost of being merge-sort-based (extra space, and no faster than $\mathbf{O}(N \log N)$ on unordered data) — but it is adaptive, so on data that is already close to sorted (few inversions, I), its cost drops towards $\mathbf{O}(N)$, which quicksort's partitioning does not exploit in the same way. HuskySort's own strategy follows directly from this contrast: husky-coding turns step 2's input into something quicksort (or, as described in § 3.2, radix sort) can sort at its best case — the coded longs are effectively unordered, cheap-to-compare data — and reserves Timsort for exactly the regime where it is strongest: cleaning up the (hopefully few) remaining inversions in step 3, where the array is already close to sorted rather than random.

Fig. 1. HuskySort Flow



3 ALGORITHM

3.1 Overview

The strategy of HuskySort (see Flow 1) is as follows (assuming that the dataset is presented as an array xs of type X): see Algorithm 3

Step 1: a Create a $long[]$ array ($longs$) of the same size as xs;
b for each element x of xs, set the corresponding element of $longs$ to $x.huskyCode()$.

Step 2: Sort array $longs$ using quicksort (in practice, we have found that Introsort[Musser 1997] works well) in such a way that when elements i, j of $longs$ are exchanged, then elements i, j of xs are also exchanged;

Step 3: Finally, when quicksort is finished and, if necessary, we run Timsort on the array xs to ensure that its elements are truly in order.

For the details of the husky-coding itself, see Algorithm 6

3.2 RadixHuskySort

Step 2 sorts the husky codes using Introsort, an $\mathbf{O}(NlogN)$ algorithm, but the codes themselves are fixed-width 64-bit values, and radix sort can sort k-bit keys in $\mathbf{O}(N \cdot \frac{k}{b})$ time using b-bit digits — linear in N for any fixed digit width b. Given that HuskySort's entire premise is to move work from the linearithmic phase to the linear phase, this raises a natural question we had not previously addressed: could step 2 itself be made linear? This section describes RadixHuskySort, which we built to answer it (Algorithm 1).

Radix sort's linear-time advantage rests on a further assumption beyond fixed key width: that a digit can be turned into an array index in $\mathbf{O}(1)$ time, so that counting and scattering never need to compare two digits directly. A b-bit digit of a husky code satisfies this trivially — it is extracted by a single shift and mask, and used directly as an index into a 2^b-bucket counting array. The original *Comparable* type generally does not: without first reducing it to such a code, radix sort applied directly to a *String*, a *Tuple*, or a case class would need some other way to decide which of 2^b buckets a given key belongs to, and whenever the natural notion of a "digit" for such a type is not already a small, dense integer, that decision falls back to *equals* (and, for a hash-based bucket structure, *hashCode*) on the type itself — an $\mathbf{O}$(size of the value) operation, not an $\mathbf{O}(1)$ one. Husky encoding is

Table 1. Prior comparison-based sorting algorithms

Algorithm	Average case	Worst case	Stable	Notes
Insertion sort	$\mathbf{O}(N + I)$	$\mathbf{O}(N^2)$	Yes	I = number of inversions; fast only for small or nearly-sorted N
Merge sort	$\mathbf{O}(N \log N)$	$\mathbf{O}(N \log N)$	Yes	Linearithmic regardless of input order; $\mathbf{O}(N)$ extra space
Quicksort	$\mathbf{O}(N \log N)$	$\mathbf{O}(N^2)$	No	In-place, cache-friendly; the quadratic worst case is mitigated in practice (e.g., Introsort's heapsort fallback[Musser 1997])
Timsort	$\mathbf{O}(N + I)$ to $\mathbf{O}(N \log N)$	$\mathbf{O}(N \log N)$	Yes	Hybrid of merge sort and insertion sort, adaptive to existing ascending/descending runs; more precisely $\mathbf{O}(N + N \log p)$ for p runs[Auger et al. 2018]

what removes this cost from every one of RadixHuskySort's $\frac{64}{b}$ digit passes, not merely from the final comparison: once the one-time $\mathbf{O}(N)$ encoding step has run, *equals* and *hashCode* on the original type are not called again until the optional cleanup pass.

A naive adaptation — swapping the object reference alongside each long at every exchange within every digit pass, exactly as step 2's Introsort swaps at every partition exchange — would incur $\mathbf{O}(N)$ reference movements per digit pass, or $\mathbf{O}(N \cdot \frac{64}{b})$ movements in total. Since $\frac{64}{b}$ is typically much smaller than *logN* at the sizes we tested, this would still usually beat Introsort's swap cost, but it needlessly reintroduces exactly the kind of payload-movement overhead this whole design exists to avoid, and least-significant-digit (LSD) radix sort's per-pass counting-sort structure makes a better option available.

Instead, RadixHuskySort's step 2 carries only an *int[]* index array through the LSD digit passes, alongside the (unmodified) long array; each pass is a stable counting sort keyed on one b-bit digit of the (already-computed) husky code, reordering the index array rather than the codes or the objects themselves. Once all $\frac{64}{b}$ passes are complete, the index array holds the sorting permutation, and a single $\mathbf{O}(N)$ pass builds the final object array by reading *xs[index[i]]* for each i. This confines all object-reference movement to one pass total, rather than one pass per digit.

Because LSD radix sort's per-digit counting sort treats keys as unsigned, and husky codes are signed 64-bit values whenever the underlying domain can be negative (e.g., *Integer*, *Long*, or dates before a chosen epoch), each code is first transformed by $code \oplus Long.MIN\_VALUE$ — flipping only the sign bit — before the digit passes, which places negative codes before non-negative ones under unsigned digit comparison without disturbing relative order within either group; the same transform is idempotent to reverse, so no separate un-flip step is needed once the permutation is built from the transformed codes. Whether a given *HuskyCoder* can actually produce negative codes is checked rather than assumed, since several codings (e.g., Unicode strings) never do.

The digit width b is a tunable parameter, trading off the number of passes ($\frac{64}{b}$) against the size of the per-pass counting array (2^b buckets, which must fit comfortably in cache for each pass to stay fast); § 6.3 reports how this trade-off plays out empirically across data types and array sizes. Step 3 (the cleanup pass, run only if the coding is not provably *perfect*) is unchanged regardless of which algorithm sorts the codes in step 2 — radix sort is a drop-in replacement for that one step, not a change to the overall three-step strategy.

This parameterization also addresses a related limitation: assuming 64-bit words throughout may not be appropriate for comparison on different key sizes. Quicksort's asymptotic behaviour does not depend on key width at all — an $\mathbf{O}(N \log N)$ comparison sort costs the same whether the keys being compared are 32 bits or 128 bits wide, since only the (fixed-cost) comparison itself touches the key's actual bits. Radix sort, by contrast, makes the key width an explicit, first-class parameter: a k-bit key simply takes $\frac{k}{b}$ passes instead of $\frac{64}{b}$, with no change to the algorithm itself. So while nothing about the discussion above depends on 64-bit words specifically, the RadixHuskySort generalizes to other key widths more directly than a comparison-based sort does — if a future encoding needed, say, 128 bits to capture more of an object's structure, the same radix sort would apply unchanged, merely with twice as many passes.

Radix sort applied to strings specifically has its own substantial, and older, literature, distinct from the general fixed-width-key radix sort described above. Three-way radix quicksort (also called multikey quicksort) and MSD (most-significant-digit) radix sort for strings were both popularized by Bentley and Sedgewick's classic treatment[Bentley and Sedgewick 1997], and burstsort[Sinha and Zobel 2004] improves cache behaviour further by building a small trie to bucket strings before sorting. These string-specialized algorithms sort variable-length strings directly, character by character, without a separate fixed-width encoding step. RadixHuskySort's contribution is different in kind, not degree: it is not a string-sorting algorithm, but a general mechanism applicable to any *Comparable* type with a 64-bit husky encoding (strings among them), at the cost of the encoding's own fixed capture window (§ 3.4 above) and a fallback cleanup pass whenever that encoding is imperfect.

That framing carries a consequence which Table 9 bears out, and which we state rather than leave to be noticed: the string rows of that table are its most variable by a wide margin. They span from 3.4x down to 1.6x, nearly the whole of its range below dates, where every other row falls between 2.1x and 3.6x. Three factors account for the spread, and they compose. The advantage grows with the cost of the type's native comparison, since that is what the encoding replaces; it grows with the exactness of the encoding, since an imperfect one is paid for by the cleanup pass; and it shrinks in the presence of algorithms specialised to the type. Dates are the favourable extreme on all three counts — a provably perfect coding, no cleanup pass at all, and no specialised competitor — and yield the largest margin in the table at 5.9x. The string rows show those same three factors varying *within* a single data type, which is the more instructive

demonstration. Chinese words are favourable on the first two counts — an expensive Unicode comparison, captured well enough by the encoding — and sit fourth of eleven, above all but three of the non-string rows. Chinese personal names are unfavourable on the second: the native comparison is more expensive still, requiring a table lookup per character per comparison, which is exactly the cost husky encoding is meant to amortise, but names of two or three characters with recurring syllables collide often enough that the cleanup pass consumes what the encoding saves, and this is the weakest result we report. English words are unremarkable on the first two counts and distinctive on the third, being the one case in this paper with a mature specialised literature ranged against it.

Strings are unfavourable on the third count in a way no other type in this paper is. Half a century of specialised string sorting exists, and a general mechanism should not be expected to beat it on its own ground. We compared RadixHuskySort against both of the classic string sorts named above, under JMH, on English and Chinese text, in natural Unicode order so that every sort compared performs the same task. Both baselines are our own implementations, each tuned so that its fallback below the partitioning cutoff allocates nothing and compares from the depth the partitioning has already established. Against three-way radix quicksort RadixHuskySort is faster at every size tested, by 1.3–2.4x on English and 3.1–4.0x on Chinese, with non-overlapping 99.9% confidence intervals throughout. Against MSD radix sort the result goes the other way at scale, though not by much: on English text the two are indistinguishable at $N = 32{,}000$, and MSD is faster by 1.20x at $N = 200{,}000$ and by 1.06x at $N = 1{,}000{,}000$. The advantage at $N = 200{,}000$ is the robust one. At $N = 1{,}000{,}000$ three runs on the machine of record give 1.00x, 1.05x and 1.06x, so at the largest size we measure the two algorithms are level to within the spread between runs, and the direction should be read from the middle size rather than the largest. We give this result because it locates the boundary rather than obscuring it. MSD earns it on a corpus that suits it and within limits that are its own: our implementation indexes an alphabet of 256 characters beyond ASCII, which English text fits and neither Chinese corpus does, and it offers no pinyin ordering, so there is no MSD row for either Chinese corpus. RadixHuskySort reaches every corpus in this paper through one encoding, with no per-alphabet provision and no per-type implementation. That is the claim being made, and the English result bounds it without contradicting it. A direct empirical comparison against burstsort remains future work; it is a trie-based, cache-conscious algorithm designed to beat both baselines used here on exactly this workload, and we have not implemented it.

Natural Unicode order is not a meaningful comparison for Chinese personal names specifically — it is simply the wrong order for that use case, not merely a different one — so a fair comparison there requires a pinyin-aware MultikeyQuicksort. We extended it with the same syllable-then-tone character key the pinyin encoding of § 3.2 already uses, restricted the comparison to the three pinyin-aware sorters (a natural-order system sort is not a real competitor for this corpus), and re-ran it. Across two independent runs, confidence intervals were wide enough (up to ±29%) that the finer RadixHuskySort-vs-QuickHuskySort distinction was not resolvable — a question the crossover analysis of § 6.5 already answers precisely on cleaner data — but one result held directionally in both attempts: RadixHuskySort and QuickHuskySort, both pinyin-aware, beat MultikeyQuicksort by roughly 1.6–2.7x, consistent with the margins measured above for English and Chinese natural-order text.

Algorithm 1: RadixHuskySort.sort

Input: *xs as an array of object X which extends Comparable<X>*
Input: *b as the number of bits per digit*

```
1 coding = huskyCoder.huskyEncode(xs);
2 longs = coding.longs;
3 index = radixSortIndices(longs, b);
4 ys = permute(xs, index);
5 if coding.perfect then
6     return ys;
7 Arrays.sort(ys);
```

Algorithm 2: radixSortIndices (LSD, deferred permutation)

Input: *longs as an array of long*
Input: *b as the number of bits per digit*
Output: *index, a permutation of* $0, \ldots,$ *longs.length* $- 1$ *such that longs[index[·]] is sorted*

```
1 biased[i] = longs[i] XOR Long.MIN_VALUE, for each i;
2 index = [0, 1, …, longs.length - 1];
3 for digit = 0 to (64 / b) - 1 do
4     index = stableCountingSortPass(biased, index, digit, b);
```

3.3 Discussion of Husky Encoding

Step 1b requires the definition of a 64-bit hash function (huskyCode) whose properties are as follows: (refer to Algorithm 4)

- If $x = y$, then $x.huskyCode() = y.huskyCode()$;
- If $x > y$, then:
  - either $x.huskyCode() <= y.huskyCode()$ (with probability p)
  - or $x.huskyCode() > y.huskyCode()$ (with probability $1 - p$);

We interpret this rule as follows: if x and y are the same ($x.equals(y)$ in Java), their huskyCode should be the same. If $x > y$, we normally expect that $x.huskyCode() > y.huskyCode()$, i.e., the sort on the husky-coded longs will not invert x and y. However, when $x > y$ while $x.huskyCode() < y.huskyCode()$, then an inversion will be introduced. And, when $x > y$ and $x.huskyCode() = y.huskyCode()$, then an inversion *may* be introduced because quick-sort is not stable.[Xiang 2011]

For a successful encoding, the value of p, the probability of a potential inversion, should be less than some critical probability, p_{crit}. Leaving a more detailed discussion of p_{crit} until later, we note that, for some domains of X, we can definitely assert that $p = 0$. We refer to this as a "perfect" encoding. In such a case, we can skip step

3 of the algorithm because, when $p = 0$, there will be no inversions remaining after quick-sorting according to the huskyCodes. An example of such a domain is date-time values with a resolution of one nano-second over a period of 128 years. For strings of English case-independent letters (no punctuation or numbers), i.e., 5-bit ASCII, words of 12 (or fewer) characters in length can also be coded uniquely. For case-dependent strings (6-bit ASCII), we can encode perfectly if the lengths are all 10 characters or less. In step 3, it might be acceptable to use insertion sort instead of Timsort. In any event, assuming that the huskyCode is as accurate as required, then there will be very few inversions (elements out of place) remaining after the quicksort phase. This implies that the second sort can be performed in very close to linear time. The results have been very satisfactory. For example, taking an array of 1,000,000 English words drawn randomly from a corpus of 275,333[Goldhahn et al. 2012], HuskySort sorts in 732 ms against the system sort's 1,194 ms, a factor of 1.63 (for the full data, please refer to table 7).

3.4 Discussion of p_{crit}

When $p > 0$, there is a finite probability of any pair of elements remaining inverted after the first sorting pass. For an array of N randomly ordered elements, the average number of inversions before sorting is $X = N(N-1)/4$. The number of inversions remaining after the first pass is simply pX. And, for Timsort or insertion sort, the time to re-sort the array will be $t = k(N + pX)$ where k is some system-dependent constant. The total time to sort the array is made up of these three time-consuming steps where k_1, k_2, k_3 are system- and algorithm-dependent constants:

Step 1: T_1 is the time to encode the elements:

$$T_1 = k_1 N$$

Step 2: T_2 is the (average) time to quick-sort the huskyCode array while also swapping the element array itself:

$$T_2 = 2k_2 N ln N$$

Step 3: T_3 is the time to Timsort the element array (only required if $p > 0$):

$$T_3 = k_3(N + pX)$$

Provided that the total time $T = T_1 + T_2 + T_3$ is less than the time required to use Timsort on the original element array, we have a win. This same three-step decomposition ($T_1/T_2/T_3$, i.e., encode/-sort/cleanup) is revisited in § 5.2 in more concrete terms — counting array accesses directly rather than folding everything into abstract system-dependent constants k_1, k_2, k_3 — and extended there to cover RadixHuskySort (§ 3.2) as a fourth alternative for the T_2 step. The actual values of p_{crit} are very much dependent on the machine running the sort. We have not calculated a typical value for p_{crit} but, even for the worst-case: Chinese characters using UTF8 encoding, we still find very significant gains, reported later in Table 7.

Of the three terms, T_3 can be isolated experimentally. Two benchmarks over the same two hundred thousand municipal records (§ 6.3) differ only in which coder they are given. Both compute identical codes; one coder declares itself perfect, so the sort skips step 3, while the other leaves that declaration at its default and the pass runs. The encoding for these records is exact, so the pass that runs has provably nothing to correct.

Its cost is predictable in advance. Timsort over a fully ascending array detects a single run and stops, performing $N - 1$ comparisons and no moves, so T_3 at $p = 0$ is exactly $N - 1$ invocations of the element's own comparator — here assessor's block, then lot, then filing date, which is precisely the comparison the husky encoding was introduced to avoid. The cleanup pass reintroduces one full composite comparison per element, having just removed it from all $\mathbf{O}(N \log N)$ of them.

That comparison costs 16.1 ns at $N = 32{,}000$, 60.3 ns at $N = 100{,}000$ and 74.2 ns at $N = 198{,}900$ — 7.3%, 17.2% and 17.3% of the running time of the sort that contains it — with non-overlapping 99.9% confidence intervals throughout. Three further runs, one of them on different hardware, agree on the shape at different levels; across all four the pass costs between a sixteenth and a quarter of the total, and between a sixth and a quarter at the two larger sizes. The percentages are the less stable figures, being ratios of two quantities that both move. The shape is a threshold rather than a trend — a rise by a factor of roughly three to four between the first two sizes, then a levelling off — and it falls where the records stop fitting in cache. That last reading is inference, but the magnitudes support it: the system sort compares these same records some 3.5 million times in 151 ms, or 43 ns each, so the cleanup pass at 74 ns costs nearly twice as much per comparison using the same comparator. Step 2 permutes references without moving the records, so a scan sequential in the array is close to random in the heap.

Whether this share must shrink as N grows — the pass being linear where the sort containing it is linearithmic — is a fair question, and § A.2 answers it.

Two conclusions follow. First, k_3 is not a constant: it varies more than fourfold across three sizes on one machine, so no single p_{crit} can be quoted even for one type on one machine. Second, an encoding which can be declared perfect avoids a cost of up to a quarter of total running time, where one which cannot must repay it before correcting a single genuine inversion.

3.5 Why HuskySort Works

So, why does this work? Well, first of all, we observe that 64-bit word length in today's computers is the norm. This allows for an encoding which has a low, possibly zero, value of p_{crit}.

Even though a Unicode string can encode only the first 3 15/16ths characters in 64 bits, this is extremely helpful in getting the elements close to their final order. The 15/16ths comes from a final unsigned right shift by one bit, applied after packing four 16-bit characters into the 64-bit code: without it, any string beginning with a character whose code point is 0x8000 or above (common among CJK characters) would set the sign bit and be encoded as a negative *long*, corrupting simple numeric comparison. The shift costs exactly the least-significant bit of a true fourth character; for strings of three characters or fewer, that bit is always an unused padding zero, so nothing is actually lost — which is exactly why this encoding is only declared *perfect* up to three characters, not four.

Here, we may compare the method of HuskySort with that of ShellSort.[Sedgewick 1996]

ShellSort acts exactly like a series of insertion sorts but, whereas each compare/swap in insertion sort is limited to resolving one

inversion only, so in ShellSort, h inversions can be resolved in one compare/swap (where we are h-sorting the array). So, Shellsort's primary mechanism is to try to fix as many inversions as possible before taking the subsequent step.

In the same way, HuskySort fixes as many inversions as possible before doing the final step. Note that it isn't necessary for the encoding to be perfect. But it should be good.

Another way of looking at it is that running quicksort or merge sort (or their variations) will require $\mathbf{O}(NlogN)$ comparisons. Merge sort, like insertion sort, is linear when the number of inversions is small and grows with $\mathbf{O}(N)$. But merge sort does use extra memory. But values such as 64-bit integers (int or long), doubles, can be compared extremely quickly in modern machines—far quicker than the time required to follow a reference (pointer) to an object in memory. In Java, this distinction is described by contrasting primitives with objects. Primitives can be quick-sorted very fast, and *long* is a primitive data type.

Exchanging (swapping), on the other hand, takes the same amount of time whatever type you are sorting because swapping an object reference is just the same as swapping any 64-bit word. The other reason why quicksort is so fast is that it is cache-friendly: its in-place partitioning scans memory sequentially rather than jumping between widely-separated locations, giving it markedly better cache behavior than, for example, heapsort[LaMarca and Ladner 1999]. Cache behavior also distinguishes quicksort variants from one another, not just quicksort from other algorithm families: dual-pivot quicksort's own performance advantage over classic single-pivot quicksort has likewise been attributed largely to cache effects[Yaroslavskiy 2009]. (LaMarca and Ladner's study also found that a naive radix sort, sorting keys directly rather than through a deferred index permutation, has relatively poor cache behavior of its own on 1990s hardware — a finding about a different radix sort design than § 3.2's, on much older hardware, so it does not bear on this paper's own radix-sort results.) So, we perform a fast quicksort on the array of longs. The only difference between the Husky version of quicksort and standard quicksort is that the swap operation must actually do two swaps: one of the longs and one of the object references. But, again, swapping is fast because elements of both arrays (i.e., both longs and refs) have good chances of being cached.

The extra time (and space) required to perform the encoding upfront, which is **O**(N), is more than offset by the faster comparisons of longs as opposed to comparisons of objects.

After the first pass (using quicksort), we can easily determine whether our encoding was perfect—in which case, there will be no remaining inversions. We do this for non-sequence classes via a simple constant-time lookup. For sequences (in practice, these are always Strings), we check each sequence's coding perfection (according to its length), and as soon as we find a sequence that cannot be encoding perfectly, we stop checking.

If the coding was perfect, we simply skip the final pass.

Given the notion of hashing that the algorithm uses, it would seem natural to call it Hash Sort. Unfortunately, that name was already taken for quite a different algorithm[Gilreath 2004]. We chose the name Husky Sort instead, after the husky, which is the mascot of the university where this work was developed.

Algorithm 3: HuskySort.sort

Input: *xs as an array of object X which extends Comparable<X>*

```
1 coding = huskyCoder.huskyEncode(xs);
2 longs = coding.longs;
3 Introsort(xs, longs, 0, longs.length, 2 * floor_lg(xs.length));
4 if coding.perfect then
5     return;
6 Arrays.sort(xs);
```

Algorithm 4: huskyEncode

Input: *xs as an array of object X*
Output: A new *Coding object*

```
1 result = new long[xs.length];
2 for i = 0 to xs.length do
3     result[i] = huskyEncode(xs[i]);
4 return new Coding(result, perfect());
```

Algorithm 5: huskyEncode (for character sequence)

Input: *xs as an array of CharSequence X*
Output: A new *Coding object*

```
1 isPerfect = true;
2 result = new long[xs.length];
3 for i = 0 to xs.length do
4     X x = xs[i];
5     if isPerfect then
6         isPerfect = perfectForLength(x.length());
7     result[i] = huskyEncode(xs[i]);
8 return new Coding(result, isPerfect);
```

Algorithm 6: stringToLong

Input:
str as String;
maxLength as the max length can be encoded;
bitWidth as how many bits will one char be encoded into;
mask as a modifier for different encoding;
Output: A new *Coding object*

```
1  length = Math.min(str.length(), maxLength);
2  padding = maxLength - length;
3  result = 0;
4  if mask == 0 then
5      for i = 0 to length do
6          result = result « bitWidth | str.charAt(i);
7  else
8      for i = 0 to length do
9          result = result « bitWidth | str.charAt(i) & mask;
10 result = result « bitWidth * padding;
11 return result;
```

4 IMPLEMENTATION

4.1 System Environment

Three machines appear in this paper, and Table 2 lists all three. Machine A ran the JDK 1.8.0_152 comparison-sort benchmarks of § 5.2, Table 5 among them; machine B, on different hardware and two JVM generations later, ran the radix-sort and JMH development work. Machine C is the machine of record: an independently administered cloud instance, confirmed idle before each run, and the source of every figure quoted in this paper. It is ARM-based like machine B but differs from it in core design, vendor, core count and operating system.

Reporting all three lets a reader check directly whether the qualitative finding generalizes across machines and JVM versions rather than being an artifact of one environment. It does: HuskySort, and now radix sort, beat the system sort on all three, which differ in vendor, instruction set, core design and JVM generation.

4.2 Implementation of Algorithm

The top-level *sort* method and the default *huskyEncode* method are implemented directly as shown in Algorithms 3 and 4 above; rather than repeat that logic here as Java source, we describe only the details not already covered by those two listings.

Coding is a simple immutable holder combining the resulting *long[]* array of codes with the *perfect* flag used by both algorithms (two fields, set once by its constructor).

unicodeToLong and *asciiToLong* are both thin wrappers around *stringToLong* (Algorithm 6), differing only in which bit-width, maximum length, and mask constant they pass in; *unicodeToLong* additionally discards the lowest bit of the packed result (an unsigned right shift by one) to guarantee a non-negative code regardless of the input string's content. Table 3 lists the constants each wrapper uses.

5 TEST CASE AND ANALYSIS

5.1 Data Source

The sources for the Strings (English and Chinese) are retrieved from Leipzig Corpora Collection[Goldhahn et al. 2012]. Chinese personal names, used for the pinyin-order comparisons of § 6.3, are a separate corpus[Wainshine 2020]. Every other dataset in this paper is generated: pseudo-random numbers, dates and tuples. The building-permit records of § 6.3 are not. They are 198,900 rows of a public municipal dataset, ordered by a composite key of assessor's block, lot and filing date, and are included because their distribution is the city's rather than ours.

In each case, we take data from the appropriate *sentences.txt* file and break it up into strings as shown in Listing 1.

```
private static String[] getLeipzigWordsFromResource(final
      String resource) {
    return getWords(resource, HuskySortBenchmark::
     getLeipzigWords);
}
private static List<String> getLeipzigWords(final String
     line) {
    return HuskySortBenchmarkHelper.splitLineIntoStrings(
     line, REGEX_LEIPZIG, REGEX_STRINGSPLITTER);
}
final static Pattern REGEX_LEIPZIG = Pattern.compile("
     [~\\t]*\\t(([\\s\\p{Punct}\\uFF0C]*\\p{L}+)*)");
public static final Pattern REGEX_STRINGSPLITTER =
     Pattern.compile("[\\s\\p{Punct}\\uFF0C]");


static List<String> splitLineIntoStrings(final String
     line, final Pattern lineMatcher, final Pattern
     stringsplitter) {
    final Matcher matcher = lineMatcher.matcher(line);
    if (matcher.find()) return Arrays.asList(
     stringsplitter.split(matcher.group(1)));
    else return new ArrayList<>();
}

static String[] getWords(final String resource, final
     Function<String, List<String>> stringListFunction) {
    try {
        File file = new File(getPathname(resource,
     DutchHuskySort.class));
        return getWordArray(file, stringListFunction, 2);
    } catch (FileNotFoundException e) {
        logger.warn("Cannot find resource: "+resource, e)
     ;
        return new String[0];
    }
}
    } catch (final Exception e) {
        throw new RuntimeException(e);
    }
}
```

Listing 1. Data Source

5.2 Analysis

Let's try to do an analysis of HuskySort's performance. This expands on the $T_1/T_2/T_3$ decomposition introduced in § 3.4 above, replacing its abstract time constants with concrete array-access counts, and extending it to cover radix sort (§ 3.2) as an alternative to quicksort for the T_2 step. The first thing to note is that, like merge sort, HuskySort requires $\mathbf{O}(N)$ extra space. This is because, in addition to the array of object references, we must have an equal-length array of longs (the huskyCode values).

Secondly, HuskySort is not stable unless using the merge sort variation. How many comparisons and swaps does HuskySort require, on average? It has been shown for dual-pivot quicksort [Nebel] that these values are (approximately):

- Comparisons: $1.9NlnN$;
- Swaps: $0.6NlnN$.

Our experiments [refer to Quicksort Analysis table] show that each comparison requires 2 array accesses (although, because one comparand is always a pivot value, this number is effectively closer to 1.) Each swap requires 4 array accesses, although again, we will be swapping elements that have already been accessed. The effective number of (normalized) array accesses is, in total, approximately 4 rather than the expected 3.8 + 2.4. That is to say, the coefficient of $NlnN$ is approximately 4. The analysis represented by the table is for dual-pivot quicksort. [1] However, in HuskySort, each swap also must swap the object references. These will not, in general,

[1]Note that, since dual-pivot quicksort is not actually implemented in the Java library for Comparable objects, we re-implemented the class for objects.

Table 2. The three benchmark environments. Machine A supplies the comparison-sort results of § 5.2; machine B the radix-sort and JMH development work; machine C, the machine of record, every figure quoted in this paper.

	A — comparison sorts	**B** — development	**C** — machine of record
Model	MacBook Pro (MacBookPro14,3)	MacBook (Apple Silicon)	AWS EC2 c7g.4xlarge
Processor	Quad-core Intel Core i7, 2.8 GHz, Hyper-Threading	Apple M1, 8 cores (4 performance + 4 efficiency)	ARM Neoverse V1 (Graviton3), 16 vCPU, 2.6 GHz
Cache	256 KB L2 per core, 6 MB L3	128 KB L1I / 64 KB L1D per core; 12 MB + 4 MB L2	—
Memory	16 GB 2133 MHz LPDDR3	16 GB unified	30 GiB
OS	—	macOS 26.5.2	Amazon Linux 2023
JVM	1.8.0_152	OpenJDK 21.0.10	OpenJDK 21.0.12 (Corretto)

Table 3. String-encoding constants

Constant	Unicode	ASCII
Bit width per character	16	7
Max length (64 / bit width)	4	9
Mask	`0xFFFF`	`0x7F`

have been pre-fetched and so the effective total normalized array accesses (A) is approximately:

$$A = 4 + 0.6 * 4 = 6.4$$

Additionally, HuskySort requires N huskyCode constructions. Each of these requires $k + 1$ array accesses where k is proportional to the number of fields contributing to the huskyCode. For a string, that will be the number of characters in the string (except that this is limited to a relatively small number according to the encoding). For ASCII strings, it is nine. For Unicode strings, it is four.

However, keep in mind that this is linear and, while it can't be completely ignored, it doesn't contribute to the overall growth of the complexity.

Encoding's cost has not simply been assumed: we timed the encoding phase in isolation, separately from the sort that follows it, to confirm it directly. At N = 1,000,000, encoding takes 52.7ms for English words (13.7% of QuickHuskySort's 383.2ms total, 21.1% of RadixHuskySort's 249.3ms total) and 311.6ms for Chinese names encoded via pinyin (29.1% of QuickHuskySort's 1070.2ms total, 40.0% of RadixHuskySort's 779.4ms total) — confirming that encoding cost, while real and non-negligible, especially for the more elaborate pinyin encoding, remains a minority of total time in every case measured, consistent with the linear-and-subordinate role it is assumed to play above. Encoding's share of the total is larger for RadixHuskySort than for QuickHuskySort in both cases, simply because radix sort's own contribution to the total is smaller — as the sort phase gets cheaper, whatever cost remains in the encoding phase necessarily becomes a proportionally larger slice of what is left, which suggests encoding cost (particularly for pinyin) is the more promising target for further optimization now that the sort phase itself is no longer the dominant cost. Once the HuskySort is complete, we will assume that there are pN remaining inversions, where p is the probability of an idiogenic (self-inflicted) inversion. Again, while we should not ignore it completely, it is linear. The number of array access for the comparisons/swaps to fix these will be $(2 + j + 4)pN$, where j is the number of fields to be compared to get a result (j is usually smaller than k).

So, the total number of array accesses (A) for Husky-sorting an array of length N is:

$$A = (6.4lnN + (2 + j + 4)p + k + 1)N$$

Timsort would be hard to analyze in the general case and so we instead analyze merge sort, which should be approximately the same number of array accesses in the average case.

For merge sort, the analysis is much simpler: the number of comparisons for a randomly ordered array of length N is $Nlog_2N$. Merge sort does not use swaps, *per se*, but instead uses copies. Assuming that we optimize the extra copying, there will be N copies to get the algorithm started and N per level. Thus, the number of copies will be $Nlog_2N$. Each of these copies requires two array accesses while each comparison requires 2 + j (described above) array accesses.

The total number of array accesses (A) for merge sort is, therefore:

$$A = (4 + j)Nlog_2N + 2N$$

Given that $log_2N = 1.44lnN$, and assuming that:

$j = 4$ corresponding to the comparison of two strings that start with the same character but differ in the second character; and

$k = 7$ and $p = 0.1$; then

the totals (A_m for merge sort and A_h for HuskySort) come to:

$$A_m = 11.5NlnN + 2N$$

$$A_h = 6.4NlnN + 9N$$

The consequence of this is that, as expected, the linear phase of HuskySort is relatively high for small N but that is soon swamped by the saving of comparisons in the linearithmic phase. The break-even point is actually a very small value of N: four. As N increases, we expect the advantage of HuskySort to increase further. This is generally borne out by the benchmarks. Please refer to table 4.

The same style of analysis extends naturally to RadixHuskySort (§ 3.2). Each of its $\frac{64}{b}$ digit passes touches every element a small constant number of times, c, to bucket it and to record its new position in the (deferred) index array — with no partitioning or pivot-comparison overhead, and critically, no logarithmic term at all. Taking $c = 4$ (the same per-exchange cost used for HuskySort's swaps above), adding the final $\mathbf{O}(N)$ pass that applies the resulting

permutation to the object array (2 array accesses per element), and the same encoding cost ($(k+1)N$) and cleanup-pass cost ($(2+j+4)pN$, unaffected by which algorithm sorts the codes in step 2), the total array accesses for RadixHuskySort (A_r) come to:

$$A_r = \left(4 \cdot \frac{64}{b} + 2 + k + 1\right) N + (2 + j + 4)pN$$

Using the same illustrative constants as above ($j = 4, k = 7, p = 0.1$) and $b = 16$ (four digit passes, near the middle of the digit-width plateau found empirically — see § 6.3):

$$A_r = 27N$$

Table 4 extends the earlier comparison with this term. At $N = 1{,}000{,}000$, $A_r \approx 27{,}000{,}000$ against $A_h \approx 101{,}000{,}000$ and $A_m \approx 161{,}000{,}000$ — a predicted advantage of roughly 3.7x over QuickHuskySort, consistent with the 2-4x speedups actually measured (§ 6.3). The model does predict one caveat: equating A_r and A_h gives a theoretical break-even around $N \approx 17$, somewhat higher than HuskySort's own break-even against merge sort ($N = 4$, above) — because radix's per-pass cost has no logarithmic term to shrink relative to at very small N. This crossover is far below every array size actually tested, so it has no practical effect on the results reported here, but it is worth stating rather than leaving the model looking like radix wins unconditionally at every N.

Since the final pass of HuskySort will be cleaning up a relatively small number of inversions, does it matter whether we use insertion sort or Timsort? It turns out that for arrays which are smaller than about 50,000 Strings, it makes barely any difference at all. Indeed, insertion sort is sometimes faster. However, as the size of the array increases beyond this value, Timsort begins to win out over insertion sort. Please refer to table 5.

6 RESULTS AND OBSERVATIONS

6.1 Benchmarks

- Comparison of sort times for HuskySort, dual-pivot quicksort, system sort. Please refer to tables 6 and 8, which carry a dual-pivot column.
- Comparison of HuskySort with system sort for numeric types (Integer, Double, Long, BigInteger, BigDecimal): Table 6 (JMH, $N = 500{,}000$, ms ± 99.9% CI).
- Comparison of HuskySort with system sort for String types (English words, Chinese words): Table 7 (JMH, ms ± 99.9% CI).
- Comparison of HuskySort with system sort for Tuples: Table 8 (JMH, ms ± 99.9% CI).

6.2 Summary

HuskySort's target application is the sorting of randomly ordered arrays of objects on the JVM (Java Virtual Machine). Let's eliminate the non-use-cases first:

- Partially-ordered arrays: that's to say, arrays with only a linear number of inversions as would be the case when adding a relatively small number of records to a large existing index. For such use cases, Timsort (the Java system sort for objects) would be more suitable.
- Arrays of primitives: in Java, a clear distinction exists between primitive types (char, byte, short, int, float, long, double) and objects. Dual-pivot quicksort (the Java system sort for primitives) is already the fastest algorithm for sorting such types.

Thus, HuskySort is best for any object array where there's no expectation of partial ordering. Typical improvements over Timsort can be read directly off Tables 6, 7 and 8 above, which give absolute times with 99.9% confidence intervals rather than summary ranges: 1.4–1.8x across the five numeric types at $N = 500{,}000$, 1.4–1.6x on English words, 2.0–2.3x on Chinese words, and 1.3–1.5x on tuples. Note that the English margin *widens* with array size rather than decaying, from 1.41x at $N = 32{,}000$ to 1.61x at $N = 1{,}000{,}000$.

6.3 Radix Sort Results

We benchmarked every data type and corpus covered above using RadixHuskySort (§ 3.2) in place of Introsort for step 2, using JMH (the Java Microbenchmark Harness, with proper fork isolation and warmup) on JDK 21.

Table 9 reports radix's advantage over QuickHuskySort as a direct multiplier — e.g., **2.9x** means radix took roughly a third as long — rather than as a percentage, since a percentage figure close to or above 100% becomes ambiguous about exactly which quantity it is a percentage *of*.

This is worth confronting directly, since a reader might reasonably ask whether either advantage erodes as N grows: a constant-factor saving on a linearithmic sort could be swamped at scale. Neither does. QuickHuskySort's margin over the system sort on English text *widens* with size, from 1.41x at $N = 32{,}000$ to 1.49x and then 1.61x (Table 7), and holds roughly constant on Chinese. Radix sort's advantage over QuickHuskySort likewise does not shrink with N: at every size and data type in Table 9, the margin is 1.5x or greater, consistent with the theoretical model of § 3.2 predicting no logarithmic term at all, so there is no analogous reason to expect the advantage to erode as N grows further. The spread across that table is itself informative, and follows the three factors of § 3.2. Chinese names sorted by pinyin order are the smallest margin measured (1.6x), because both QuickHuskySort and RadixHuskySort must pay the same cleanup pass for a coding that two or three characters of recurring syllables leave badly imperfect. Dates show the largest margin (5.9x) because that coding is provably perfect and never needs a cleanup pass at all, letting radix's linear-time advantage show through with nothing else in the way. The two lie at opposite ends of the same axis: what separates them is not the data type but the exactness of the encoding.

The building permits row is the only one in that table not measured on data we generated: 198,900 real municipal records ordered by a composite key of three fields (§ 5.1). At 2.1x it is tenth of eleven, which is the point worth making about it — it is not our largest margin, but it is our largest on data whose distribution we did not choose. § A.3 gives the case study, and § A.4 what a general composite encoder would have to provide, that one having been written by hand.

The measurements were reproduced qualitatively on machines A and B of Table 2, which differ from the machine of record and from

Table 4. Theoretical comparisons (array accesses, dimensionless counts)

N	N ln N	merge sort	HuskySort	Radix
4	6	72	72	108
1,000	6,908	81,439	55,075	27,000
1,000,000	13,815,511	160,878,371	101,149,455	27,000,000
1,000,000,000	20,723,265,837	240,317,557,125	147,224,183,132	27,000,000,000

Table 5. Timsort or insertion sort as the cleanup pass (time in milliseconds)

N	Husky with Tim (ms)	Husky with Insertion (ms)
1000	0.14	0.14
2000	0.31	0.30
4000	0.63	0.64
8000	1.46	1.56
16000	3.24	3.19
32000	6.56	6.48
64000	16.85	19.41
125000	47.74	54.90
250000	168.21	167.90
500000	259.37	373.43
1000000	602.08	1202.61
2000000	1234.90	3129.23
4000000	2423.62	11018.19

Table 6. Comparison of HuskySort with system sort (Numeric, $N = 500{,}000$, ms, mean ± 99.9% CI)

Type	System	QuickHuskySort	DualPivotQuicksort	quicksort (raw)	Radix/8	Radix/16
Integer	121.2±0.6	89.7±3.5	82.5±0.2	38.1±0.1	31.2±0.5	25.2±0.2
Double	141.7±1.3	97.8±3.6	89.6±0.5	42.6±0.2	34.0±0.5	31.0±0.3
Long	135.8±1.2	97.7±1.0	89.2±0.6	38.2±0.3	30.2±0.2	28.6±0.3
BigInteger	208.4±1.7	152.3±1.7	209.9±2.0	42.2±0.1	56.9±0.3	54.1±0.2
BigDecimal	262.2±1.6	144.1±2.1	246.2±3.7	47.6±0.0	54.1±0.3	52.5±0.3

Table 7. Comparison of HuskySort with system sort (Strings, ms, mean ± 99.9% CI)

N	Corpus	System sort	QuickHuskySort	Radix/8	Radix/16
32,000	English	13.69±0.19	9.72±0.12	4.63±0.16	4.18±0.16
32,000	Chinese	9.78±0.08	5.00±0.04	2.10±0.03	1.89±0.02
200,000	English	141.39±2.59	94.74±2.44	55.01±2.20	50.27±2.51
200,000	Chinese	71.34±0.63	30.90±0.13	13.49±0.15	10.85±0.10
1,000,000	English	1120.87±14.43	697.88±16.15	355.26±24.71	272.74±6.46
1,000,000	Chinese	447.12±2.28	219.80±1.99	83.25±2.31	65.46±1.47

Table 8. Comparison of HuskySort with system sort (Tuples, ms, mean ± 99.9% CI)

N	System	QuickHuskySort	DualPivotQuicksort	Radix/16
20,000	3.72±0.01	2.79±0.02	3.41±0.09	1.30±0.01
100,000	23.24±0.19	17.65±0.16	21.36±0.48	6.82±0.05
500,000	209.53±7.98	142.61±1.83	142.70±4.98	57.98±0.76

Table 9. Radix sort advantage over QuickHuskySort

Data Type	N	Advantage over QuickHuskySort
`Dates`	20,000	5.9x
`Long`	500,000	3.6x
`Integer`	500,000	3.6x
`Strings of Chinese words (natural order)`	1,000,000	3.4x
`Double`	500,000	3.2x
`BigInteger`	500,000	2.8x
`BigDecimal`	500,000	2.8x
`Strings of English words`	1,000,000	2.6x
`Tuples`	500,000	2.5x
`Building permits (composite key)`	198,900	2.1x
`Strings of Chinese names (pinyin order)`	1,000,000	1.6x

each other in vendor, instruction set, core design and JVM generation. We quote no figures from them, since mixing environments within a comparison would rob it of meaning, but every ordering reported here holds on all three.

6.4 Parallel Radix Sort

Radix sort's digit passes are natural candidates for parallelization (§ 3.2 above). We implemented and benchmarked a parallel variant: each pass is split into a configurable number of contiguous chunks, one per worker thread. Every chunk first computes its own local per-bucket histogram independently, with no synchronization required; a short sequential step then combines those histograms into an exact starting offset, in the output array, for every (chunk, bucket) pair — preserving the stability that step 3's cleanup pass, when needed, relies on, since a chunk's elements always land after all lower-numbered chunks' elements sharing the same bucket; finally, every chunk scatters its own elements into the output arrays independently, using only its own precomputed offsets, so no two chunks ever write to the same location.

An initial implementation dispatched the two phases above to a thread pool separately on every pass. Measurement showed this repeated dispatch itself contributing a real, non-trivial share of the total cost, particularly at more moderate array sizes. The overhead was substantially reduced by starting the worker threads once per sort, rather than once per phase per pass: each thread runs its own loop over every pass, and the two phases synchronize via reused barriers whose associated actions — code that runs exactly once per synchronization point, in whichever thread arrives last — perform the histogram-combine and buffer-swap steps. Table 10 reports the result (*Long[]*, 11-bit digits, milliseconds, mean ± 99.9% CI).

Isolating parallelism itself, that's to say comparing 1 thread against 4, both paying identical thread-coordination overhead, gives a statistically resolved speedup at both sizes: 1.39x at $N = 2{,}000{,}000$ and 1.29x at $N = 10{,}000{,}000$, with non-overlapping confidence intervals in both cases. Against the serial implementation the improvement is resolved at both sizes too, 172.0ms to 118.5ms at $N = 2{,}000{,}000$ and 935.2ms to 687.9ms at $N = 10{,}000{,}000$. Unlike on the eight-core machine of an earlier draft, scaling does not stop at four threads: on sixteen uniform cores, eight threads beats four at both sizes with intervals that do not overlap, reaching 1.51x and 1.37x over a single thread.

That is still well short of linear, and the shortfall is the more interesting number: an eightfold increase in threads buys between 1.4x and 1.5x, on hardware with no performance/efficiency asymmetry to account for it. Two explanations fit, and we cannot separate them without hardware counters. Each digit pass scatters across the whole array, so the sort may simply be limited by memory throughput, which extra cores do not supply; and each pass's histogram-combine step is sequential, running once per synchronization point whatever the thread count, so it grows as a share of the total as threads are added. The first would be a property of the machine and the second of the design, and only the second would survive better hardware.

6.5 Use-Case Guidance

§ 6.2 already eliminates two cases outright: partially-ordered arrays, where Timsort's own adaptivity wins, and arrays of already-cheap-to-compare primitives, where dual-pivot quicksort is already the fastest option. For the remaining case — arrays of objects with no expectation of partial ordering — the decision is governed by the three factors of § 3.2 rather than by the size of the array, and Table 11 puts them in the order a practitioner can answer them. Size enters only once those are settled, and then only near the bottom of the range.

Only when all of those are answered does array size matter, and then chiefly at the small end, where each algorithm's fixed setup cost has not yet been amortized. For String keys we measured every crossover directly (JMH, English corpus, $N = 4$ through 10,000). System sort is the fastest of the group up to $N = 20$, though only narrowly there (0.462μs against insertion sort's 0.489μs). A plain, binary-search-based insertion sort wins outright from $N = 50$ through $N = 200$ — the one regime in this paper where plain insertion sort, not either HuskySort variant, is the best available choice. QuickHuskySort takes over between $N = 200$ and $N = 500$, and remains the best choice up to roughly $N = 2{,}000$, since RadixHuskySort's own fixed setup cost is not yet amortized below that point. That cost can be read directly off the measurements: RadixHuskySort's time is a near-constant 166μs floor from $N = 4$ to $N = 500$, the allocation and initialisation of its counting structures, which only

Table 10. Parallel radix sort: serial vs. 1, 2, 4, and 8 worker threads

N	Serial	1 thread	2 threads	4 threads	8 threads
2,000,000	172.0±5.7	165.2±2.2	140.3±2.0	118.5±2.4	109.3±1.8
10,000,000	935.2±36.9	884.5±21.6	731.6±19.9	687.9±22.1	645.3±20.9

Table 11. Which sort to reach for. The situations are ordered by how much they decide, and array size settles nothing until all of them are.

Your situation	Reach for	Evidence
The ordering is already cheap to evaluate — primitives, or a single numeric field	dual-pivot quicksort	cost of encoding cannot be recouped (§ 6.2)
The ordering is composite or expensive, *and* the key packs exactly into 64 bits	RadixHuskySort; no cleanup pass is required	the largest margins we measure: dates 5.9x, building permits 2.1x over QuickHuskySort
The ordering is composite or expensive, but the encoding is imperfect	RadixHuskySort, even allowing for the cleanup pass	that pass costs a sixth to a quarter past $N = 100{,}000$, even with nothing to correct (§ 3.4)
A sort specialised to your type already exists	use it, if it covers your data	on English words a tuned MSD radix sort beats us, though our MSD indexes extended ASCII only and cannot sort either Chinese corpus (§ 3.2)
The data defeats the encoding's capture window — long shared prefixes, or keys wider than the coder can see	System sort	every husky variant loses its advantage (§ A.5)
None of the above settles it, or the keys are strings	by array size (§ 6.5)	each sort's fixed setup cost is repaid at a different N

the linear term begins to dominate past a few thousand elements. Somewhere between $N = 2{,}000$ and $N = 10{,}000$ it takes over; its advantage then grows and holds at every larger size tested (Table 9). Every one of these crossovers shares the same shape: each successive algorithm carries a larger fixed setup cost that only pays for itself once there is enough work to amortize it against — the same fixed-vs-variable-cost story § 6.4 found one level further down, where a parallel radix sort's own thread/barrier setup only pays off once there is enough work to divide across threads. These crossover points were measured only for String keys. The fixed costs involved are architectural rather than type-specific, so other key types plausibly show similarly-shaped curves at similar sizes, though that has not been separately confirmed.

Two further caveats from elsewhere in this paper complete the picture. If the source data defeats the husky encoding's fixed capture window entirely — a shared prefix longer than the encoding's own character limit, for instance — neither QuickHuskySort nor RadixHuskySort helps regardless of N (§ A.5): both lose to plain System sort, since the encoding pass becomes pure waste and the mandatory cleanup pass must do all the real work anyway. And given a large enough workload and spare cores, *ParallelRadixHuskySort* widens RadixHuskySort's own advantage further still (§ 6.4), provided there is enough total work to amortize its own thread-coordination cost.

7 CONCLUSION

The original idea behind HuskySort, that's to say QuickHuskySort, was to shift work from the linearithmic phase of sorting to the linear phase (both the prelude and the postlude) and effect an overall reduction in array accesses and thus processing time. This has been demonstrated such that, when sorting objects rather than primitives, HuskySort is faster than dual-pivot quicksort wherever the ordering is expensive to evaluate — by 1.4x on *BigInteger* and 1.7x on *BigDecimal* (Table 6). On *Integer*, *Double* and *Long*, where a comparison costs almost nothing, the encoding has nothing to amortise and dual-pivot quicksort is ahead instead, by up to a tenth. That is the same criterion which governs every other result in this paper, here separating one group of numeric types from another. It is faster than Timsort for arrays which are not partially ordered, above a few hundred elements; below that, Timsort's own tuned handling of small arrays wins instead. It is especially fast for types whose ordering is composite or expensive to evaluate and whose keys encode exactly, where the encoding replaces the whole comparison and no cleanup pass is needed. Strings are not reliably that case: they are the one domain with a mature specialised literature of its own (§ 3.2), and they are also the most variable, spanning nearly the whole range of our results — including the narrowest margin we report, for Chinese personal names in pinyin order.

In the case of RadixHuskySort, the performance improvement is even better for datasets above a few thousand in length. Of course, in this case, there was no linearithmic phase, so the time savings

come, as explained above, from more bit manipulation and fewer calls to *equals*.

These findings are not an artifact of a single development machine. The same qualitative pattern held on three environments spanning two CPU vendors and three distinct architectures (Table 2), the last a 16-core ARM server instance with no performance/efficiency core split to confound the result. That same run also settled a question the parallel variant's own results had left open (§ 6.4): its scaling ceiling past a handful of threads is a property of the design itself, the fixed sequential cost of combining per-thread histograms once per pass, not an artifact of any one machine's particular mix of core types.

HuskySort's advantage is specific to a particular kind of workload rather than universal, and needs all three of the following to hold: an array of at least a few hundred elements, growing to a few thousand before RadixHuskySort's own per-pass setup cost is repaid (§ 6.5); no expectation of pre-existing order, which favours Timsort's adaptivity instead (§ 6.2); and a comparison costing more than a single machine-word check — strings, arbitrary-precision numbers and multi-field records, as opposed to *int*, *long* or *double*, where dual-pivot quicksort already wins outright.

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A APPENDIX

A.1 Discussion of coding accuracy

It is not essential that coding is perfect for HuskySort to work well. We have experimented with deliberately throwing off the encoding quite drastically. If the probability of one of these deliberate miscodings is less than approximately 25 percent, HuskySort can still save time. The inference that we can take from this is that it is not critical that encodings are all very good. We can be confident that we are still saving time up to about one in four errors.

These deliberate errors were tested for Bytes and Integers but not the other generic types.

A.2 The cleanup pass as N grows

§ 3.4 measures the cleanup pass at $p = 0$ and reports a share of total running time that *rises* with N, from 7.3% to 17.3%. One objection should be met head on: the pass is linear where the sort containing it is linearithmic, so its share is bound to fall as $1/\log N$.

Both are true, because k_3 is not settled over this range. It grows more than fourfold between the smallest and largest sizes measured while $\ln N$ grows by a fifth, so the cache effect swamps the logarithm completely. Holding k_3 at the value it reaches once the records no longer fit in cache, and growing the rest as $\ln N$, puts the share at some 16% at $N = 10^6$ and 14% at $N = 10^7$: the decline is real but logarithmic, and the pass does not become negligible at any size we expect to meet.

That last calculation is a model rather than a measurement. The sort's own per-element cost also grows faster than $\ln N$ across these three sizes, and the corpus holds 198,900 records, so the measurement cannot be extended on real data.

A.3 Building permits: the case study in full

The permits row of Table 9 is the case this mechanism is designed for. The ordering is expensive, since a comparison may have to examine all three fields of the key in turn, and the key nevertheless packs exactly into 60 bits, so no cleanup pass is required at all. Against the system sort, RadixHuskySort is faster by 5.17x, 4.26x and 4.53x at N = 32,000, 100,000 and 198,900; against QuickHuskySort, by 2.52x, 2.06x and 2.12x.

At 2.1x over QuickHuskySort the row is tenth of eleven. A composite key of three fields is more expensive to compare than a single *Long* but a good deal cheaper than a pinyin lookup per character, so the margin falls squarely inside the range the three factors of § 3.2 predict for it. That is the point worth making about it: it is not our largest margin, but it is our largest on data whose distribution we did not choose, and a mechanism that only worked on synthetic inputs would not be worth reporting.

The same records also supply the cleanup-pass measurement of § 3.4, because the encoding for them is exact and can be declared so — which is what allows the pass to be switched on and off over identical codes.

A.4 Future work: composite keys, and what a library would have to provide

The permit coder above was written by hand, as was every composite encoding in this paper. One practical obstacle stands between that and a library implementation of this mechanism, and removing it is the piece of future work we think matters most for adoption. A hand-written coder must satisfy three properties that nothing checks. Fields must be packed most-significant-first in the same order as the type's own comparison. Symbol codes must run from one upwards, so that a field shorter than its width pads with zero and sorts low, as a string prefix does. And a symbol outside a field's alphabet must collapse to the largest symbol at or below it, so that an unexpected value weakens the ordering rather than inverting it.

The first of those is the one that bites, and it bites hardest where one might expect it to matter least. An encoding declared perfect skips the cleanup pass entirely (§ 3.4), so if its field order does not match the comparison, the sort returns the wrong answer and nothing downstream detects it. The same mistake in an imperfect encoding is merely slow: the cleanup pass re-sorts by the real comparison and the answer is still correct. Field order therefore matters most exactly when the encoding is at its best.

We expect most of this can be automated rather than merely assisted. Where a type's ordering is already derived from its field declarations — as with Java records, or Python dataclasses declared with an ordering — the encoding can be derived from the same declaration, which removes by construction the possibility that packing order and comparison order disagree, and would let the perfection flag be computed rather than asserted and separately verified. What such a derivation additionally needs is each field's range rather than its type, since it is the narrowness of the fields that lets a composite key fit at all: the permit block field is five characters of a thirty-one symbol alphabet, not an unbounded string. That range is either declared or inferred from the data.

This argues specifically for the radix variant. Once the fields do not fit in 64 bits — three fit comfortably, six will not — a wider code costs RadixHuskySort proportionally more digit passes and nothing else (§ 3.2), where a comparison-based husky sort has no comparable escape.

A.5 Adversarial inputs

The discussion in section A.1 raises a concern: it may be possible to construct inputs on which the husky encoding has poor entropy, and it is not obvious in advance how gracefully (or badly) the sort that follows would degrade in that case. We constructed two such inputs deliberately, to see how the two variants (QuickHuskySort and RadixHuskySort, § 3.2) each degrade.

Collapsed high-order bits. We generated arrays of *Long* values with a swept number of high-order bits held identical across every element (the rest random) — 0 fixed bits is the random baseline, 63 leaves only the sign bit free — and compared against a re-implementation from scratch of dual-pivot quicksort (the same baseline used for the numeric comparisons above, without any three-way or equal-key handling). That baseline is reported twice, and the pair is the most informative thing in this section. Java's own *DualPivotQuicksort* bounds its recursion at 64 levels and falls back to heapsort beyond it; our transcription of the algorithm originally did not. Table 12 reports both, at $N = 1{,}000{,}000$.

Radix sort's timings stay essentially flat across the entire sweep, exactly as the model of § 3.2 predicts: each digit pass does the same fixed amount of work regardless of how the keys are distributed, so a pass over collapsed digits (everything lands in one bucket) costs no more than any other pass.

The unguarded baseline degrades by more than an order of magnitude at 56 fixed bits and then fails outright, exhausting the stack at 60 and 63. This is a well-known failure mode for a quicksort that partitions naïvely around a heavily-duplicated pivot value: Bentley and McIlroy's classic treatment of engineering a practical quicksort[Bentley and McIlroy 1993] specifically addresses this case (a solution to Dijkstra's Dutch National Flag problem, partitioning into three regions — less than, equal to, and greater than the pivot — rather than two), precisely to avoid the pathological recursion depth this baseline exhibits. The degradation is far worse than the crash makes it look, because the crash truncates it. Removing the stack limit rather than the recursion, and timing the same input at $N = 1{,}000{,}000$ with the depth bound raised until it never fires, gives 363 seconds at 63 fixed bits against 0.4 seconds at none — a factor of nine hundred, on data of identical size.

The guarded column is the same algorithm with the bound the JDK ships, and it is a different story entirely: nothing crashes, and the 21x degradation at 56 fixed bits becomes 2.1x. What that column is measuring, however, is largely *not* quicksort. On these inputs the partitions go unbalanced immediately, the bound is reached within the first few dozen levels, and heapsort finishes the work. The clearest evidence is the direction of the sweep: unguarded, the worst case is 63 fixed bits, the most degenerate input, which is what the partitioning pathology predicts; guarded, 63 is the *fastest* cell in the row. The guard has not made dual-pivot quicksort robust here. It has detected that dual-pivot quicksort cannot cope and changed algorithm, which is what a production sort should do and is worth stating plainly, since a reader may otherwise take the guarded column for evidence that the pathology is mild.

QuickHuskySort neither crashes nor slows down anywhere in this sweep — if anything it gets faster as duplication increases, the same pattern System sort shows, consistent with Timsort's adaptive run-detection treating long runs of nearly equal keys as already sorted. Indeed both overtake radix sort at the extreme of the sweep, where 63 fixed bits leave almost nothing to sort and an adaptive algorithm has almost nothing to do, while radix still pays for its

Table 12. Timings (ms) as high-order bits are progressively collapsed, $N = 1{,}000{,}000$. The dual-pivot baseline is shown both without a recursion-depth bound and with the one the JDK itself applies.

Fixed high bits	System sort	QuickHuskySort	Dual-pivot quicksort		Radix/16
			unguarded	guarded	
0	477.6	420.0	335.7	335.9	74.8
32	493.8	420.8	340.9	331.2	69.6
48	485.1	357.9	354.6	333.3	66.8
56	174.3	197.8	**7097.7**	717.2	58.9
60	115.9	69.8	*stack overflow*	805.5	42.6
63	42.6	15.4	*stack overflow*	186.9	34.9

fixed number of passes. Radix sort's robustness is of a third kind, and the most useful of the three: it needs neither an adaptive algorithm's luck with the input nor a fallback to a different algorithm, because its cost does not depend on the distribution at all. It is the only sort here whose worst case can be read off its best.

Shared string prefixes. We took real corpus words and prepended a fixed-length prefix of repeated characters, long enough in the worst case to exceed the English coder's character-capture window (nine 7-bit ASCII characters per 64-bit code). Table 13 reports timings at $N = 1{,}000{,}000$.

This is a genuinely different failure mode, and a less flattering one. Once the shared prefix reaches the coder's nine-character window, every element's husky code becomes bit-for-bit identical — the encoding provides zero disambiguation — and *every* Husky-based approach, radix included, loses its advantage completely, finishing level with plain System sort or a little behind it rather than ahead. The mechanism has nothing to do with which algorithm sorts the codes in step 2: once the encoding carries no information at all, the entire ordering burden falls on the mandatory cleanup pass (step 3) doing real string comparisons — comparisons that are themselves more expensive than usual, since they must scan past the shared prefix before finding a difference — and whichever algorithm ran first in step 2 did genuinely useless work before that. Radix does not recover any advantage here, and is very slightly the slowest of the three Husky-based options in this regime, since its fixed per-pass cost (a strength above) becomes pure overhead once there is no information left to sort.

Taken together, these two experiments answer the question posed above with two distinct results. When keys collide in their high-order bits but the encoding itself is still informative, radix sort is effectively immune, and a comparison sort is not: unguarded it degrades by orders of magnitude or fails outright, and guarded it survives only by abandoning the algorithm one asked for. But when the source data overwhelms the encoding's fixed capture window entirely, no choice of sort algorithm for the encoded longs helps at all; this is a limitation of the fixed-width encoding scheme itself, not of HuskySort's algorithmic strategy, and it is already implicit in the p_{crit} discussion of § 3.4: whenever the actual input defeats the encoding, the result is that no downstream sort algorithm can recover the information the encoding failed to capture.

Worth noting as a related, if incidental, finding: the three-way radix quicksort baseline used in § 3.2 for the classic string-sorting-literature comparison (MultikeyQuicksort) shares exactly the same failure mode as the naive quicksort baseline above, for a similar structural reason. Its equal-partition recursion advances one level per shared prefix character; on a long enough run of strings sharing a common prefix — the same kind of adversarial construction as above — a straightforward recursive implementation consumes one stack frame per shared character and eventually overflows the stack. Unlike the failure discussed above, this is a property of the comparison algorithm itself and unrelated to Husky encoding. We fixed it in our own implementation by converting that one recursive call into a loop, restoring constant stack depth regardless of prefix length.

The MSD radix sort baseline needs a comparable fix, for a different reason. A run of equal strings occupies a single bucket at every depth beyond their length, so recursion into that bucket reduces neither the range nor the work outstanding. The cutoff to insertion sort conceals this below its own threshold, and a corpus sampled with replacement exceeds that threshold at scale. All three of the comparison baselines discussed here therefore carried an unbounded-recursion failure mode, and all three have now been repaired: the multikey recursion converted to a loop, the MSD cutoff corrected, and the dual-pivot baseline given the depth bound the JDK applies. Only the third of those changes what the sweep reports, which is why it is shown both ways. RadixHuskySort was never at risk of any of them in the first place, since LSD radix sort has no recursion to begin with.

A.6 Handling of partially-sorted arrays

Not surprisingly, HuskySort is best suited for sorting random arrays. The benefits of partially-sorted arrays are enjoyed most by the merge sort family of algorithms (including Timsort). However, we have studied the efficacy of a modified merge sort (it is required to manage both the objects and the longs) as the first pass of the HuskySort algorithm. For random English String inputs, the merge-sort variation improves on system sort by approximately 20 percent whereas the normal intro-sort-based HuskySort improves on the system sort by more like 40 Percent.

However, it is not recommended to use HuskySort when arrays are partially ordered. In such cases, the merge-sort variation of HuskySort will take anywhere between one and one half times and four times as long as Timsort.

Table 13. Timings (ms) as a shared string prefix grows, $N = 1{,}000{,}000$

Prefix length	System sort	QuickHuskySort	Radix/16
0	1351.6	583.4	234.4
10	862.1	867.0	873.0
20	861.7	912.7	904.1
40	887.6	944.0	967.4